

- No overlap of time between each trip of the same vehicle.
- Each customer is served exactly once by one vehicle.
- The total demand served on the trip of a vehicle cannot exceed the vehicle's capacity.

3.2. Model formulation

In this section, we formulate the Multi-D&T VRPTW-R as a MIP model.

3.2.1. Notations

Indices and sets:

i, j, l	index of a customer.
N	set of customers.
k	index of a vehicle.
K	set of vehicles.
w	index of a trip.
W	set of trips for any vehicle.

Note: For any vehicle, the number of the real performed trip(s) gradually increases from 1 with a step size of 1. For instance, for two vehicles k and k' with two and three trips, respectively, the real performed trip sets are $\{1,2\}$ and $\{1,2,3\}$ respectively.

Parameters:

e_i	the earliest service time at customer i .
l_i	the latest service time at customer i .
q_i	the demand for customer i .
t_i^R	release date of packages requested by customer i .
t_i^S	service time of customer i .
t_{ij}^N	travel time between customer i and j .
t_k^D	service time of vehicle k 's depot.
$t_{k,i}$	travel time between vehicle k 's depot and customer i .
T^H	time horizon.
Q	the capacity of each vehicle.
M	a sufficiently large positive number.

Decision variables:

$\tau_{k,w}^L$	time point that vehicle k departs from the depot in trip w .
$\tau_{k,w}^B$	time point that vehicle k returns to the depot in trip w .
$\tau_{k,i,w}$	time point that vehicle k arrives at customer i in trip w .
$\delta_{k,w}^L$	cargo load of vehicle k in trip w when departing from the depot.
$\delta_{k,i,w}$	cargo load of vehicle k in trip w after vehicle k visiting customer i .
$\gamma_{k,w}$	binary equals one if vehicle k operates trip w ; otherwise, equals zero.
$\xi_{i,j,k,w}$	binary equals one if vehicle k visits customer j immediately after visiting customer i in trip w ; otherwise, equals zero.
$\lambda_{k,i,w}$	binary equals one if vehicle k serves customer i in trip w ; otherwise, equals zero.
$\mu_{k,i,w}$	binary equals one if customer i is the first customer served by vehicle k in trip w ; otherwise, equals zero.
$\eta_{k,i,w}$	binary equals one if customer i is the last customer served by vehicle k in trip w ; otherwise, equals zero.

3.2.2. Mathematical model

$$\text{Minimize } Z = \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} \sum_{j \in N} t_{ij}^N \xi_{i,j,k,w} + \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} t_{k,i} \mu_{k,i,w} + \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} t_{k,i} \eta_{k,i,w} \quad (1)$$

Subject to:

$$\sum_{k \in K} \sum_{w \in W} \lambda_{k,i,w} = 1 \quad \forall i \in N \quad (2)$$

$$\mu_{k,l,w} + \sum_{i \in N} \xi_{i,l,k,w} = \eta_{k,l,w} + \sum_{j \in N} \xi_{l,j,k,w} = \lambda_{k,l,w} \quad \forall l \in N, k \in K, w \in W \quad (3)$$

$$\sum_{i \in N} \mu_{k,i,w} = \sum_{i \in N} \eta_{k,i,w} = \gamma_{k,w} \quad \forall k \in K, w \in W \quad (4)$$

$$\gamma_{k,w+1} \leq \gamma_{k,w} \quad \forall k \in K, w \in W/|W| \quad (5)$$

$$\sum_{i \in N} \lambda_{k,i,w} \leq \gamma_{k,w} |N| \quad \forall k \in K, w \in W \quad (6)$$

$$\delta_{k,w}^L = \sum_{i \in N} \lambda_{k,i,w} q_i \quad \forall k \in K, w \in W \quad (7)$$

$$\delta_{k,w}^L \leq \gamma_{k,w} Q \quad \forall k \in K, w \in W \quad (8)$$

$$\tau_{k,i,w} \leq \lambda_{k,i,w} l_i \quad \forall i \in N, k \in K, w \in W \quad (9)$$

$$\tau_{k,w}^L \geq \lambda_{k,i,w} (t_i^R + t_k^D) - M(1 - \gamma_{k,w}) \quad \forall i \in N, k \in K, w \in W \quad (10)$$

$$\tau_{k,w+1}^L \geq \tau_{k,w}^B + t_k^D - M(1 - \gamma_{k,w+1}) \quad \forall i \in N, k \in K, w \in W \quad (11)$$

$$\tau_{k,i,w} \geq \tau_{k,w}^L + t_{k,i} - M(1 - \mu_{k,i,w}) \quad \forall i \in N, k \in K, w \in W \quad (12)$$

$$\tau_{k,w}^B \geq \max\{\tau_{k,i,w}, e_i\} + t_i^S + t_{k,i} - M(1 - \eta_{k,i,w}) \quad \forall i \in N, k \in K, w \in W \quad (13)$$

$$\tau_{k,j,w} \geq \max\{\tau_{k,i,w}, e_i\} + t_i^S + t_{i,j}^N - M(1 - \xi_{i,j,k,w}) \quad \forall i, j \in N, k \in K, w \in W \quad (14)$$

$$\tau_{k,w}^B \leq T^H \quad \forall k \in K, w \in W \quad (15)$$

$$\gamma_{k,w} \in \{0, 1\} \quad \forall k \in K, w \in W \quad (16)$$

$$\xi_{i,j,k,w} \in \{0, 1\} \quad \forall i, j \in N, k \in K, w \in W \quad (17)$$

$$\lambda_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (18)$$

$$\mu_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (19)$$

$$\eta_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (20)$$

The objective function (1) minimizes the total traveling time of all the vehicles. Constraints (2) impose that each customer is served exactly once. Constraints (3) show that, for a customer who is served in a trip of a vehicle, if he/she is the first customer in the served customer sequence of this trip, then no customer can be served before him/her in this trip, otherwise, there must exist a customer who is served exactly before him/her. Similarly, if he/she is the last customer in the served customer sequence of this trip, then no customer can be served after him/her in this trip, otherwise, there must be a customer who is served exactly after him/her. Constraints (4) ensure that there must exist only one first customer point and one last customer point in a trip that is performed, on the contrary, if the trip is not performed, then no first customer point exists in this trip, nor the last customer point. Constraints (5) guarantee that trip $w + 1$ of vehicle k can be operated only if vehicle k has operated trip w . Constraints (6) guarantee that a vehicle can serve a customer only if the customer is included in this vehicle's trip. Constraints (7) ensure that the total loads of a vehicle equal the sum of customer demands. Constraints (8) ensure that the total loads of a vehicle cannot exceed its vehicle capacity. Constraints (9) are the service time window constraints. Constraints (10) address the time relationship between the departure time of a vehicle and release dates as well as depot service time in a trip. Constraints (11) state the time relationship between consecutive trips of the same vehicle. Constraints (12)–(13) are travel time conservation constraints. Constraints (14) are service time conservation constraints. Constraints (15) ensure that the time that a vehicle is returned to the depot is no later than the time horizon T^H . Constraints (16)–(20) define the domains of decision variables.

It should be noted that constraints (11), (13), (14) are presented in a nonlinear form with max functions, there is no necessity to linearize them because the current version of the model implementation tools applied in this study (introduced in Section 5) already support the max/min functions.

4. Algorithmic strategies

The model developed in Section 3 can be solved by some off-the-shelf solvers (e.g., CPLEX) for small-scale instances. To deal with large-scale instances, we develop an HPSO algorithm and an HGA algorithm to solve our proposed model and compare their performance through extensive experiments. The classical particle swarm optimization (PSO) algorithm initializes a set of positions of particles (i.e., the swarm), and obtains the optimal position by iteratively updating the positions towards the incumbent best one. In our strategies, HPSO combines the local search-variable neighborhood descent (LS-VND) and the solution translation mechanism to expand search neighborhoods as well as accelerate the algorithm. The classical genetic algorithm (GA) initializes a set of chromosomes (i.e., the population) and the best individual information is found through evolution and iteration mechanisms. In our strategies, a reorder routine is adapted to accelerate our HGA.

In Section 4.1 and 4.2, we separately describe the frameworks for the proposed HPSO and HGA. There are several same procedures in HPSO and HGA, such as LS-VND, labeling procedure. The details of these procedures are all introduced as follows.

4.1. HPSO algorithm

In HPSO, particles in swarm represent solutions. Each particle is identified by a customer id sequence (sorted by increasing order) and a corresponding *random position* sequence (see Fig. 3(a)). Then, a labeling procedure is executed to evaluate the fitness value of